

Task 1: Build and Evaluate a Linear Regression Model for House Price Prediction

1. Introduction

Machine Learning is a branch of Artificial Intelligence that enables computers to learn patterns from data and make predictions. In this task, a Linear Regression model was developed to predict house prices using the California Housing Dataset provided by Scikit-Learn.

The objective of this project was to understand the basic machine learning workflow, including data loading, data exploration, model training, prediction, and performance evaluation.

2. Dataset Description

The California Housing Dataset contains information collected from different districts in California. The dataset includes several features related to housing and population statistics.

Some important features are:

- MedInc – Median income in the area
- HouseAge – Average age of houses
- AveRooms – Average number of rooms
- AveBedrms – Average number of bedrooms
- Population – Population of the area
- AveOccup – Average house occupancy
- Latitude and Longitude – Geographic location

The target variable is:

- MedHouseVal – Median house value

The dataset contains 20,640 records and 8 input features.

3. Exploratory Data Analysis (EDA)

Before training the model, the dataset was explored to understand its structure and characteristics.

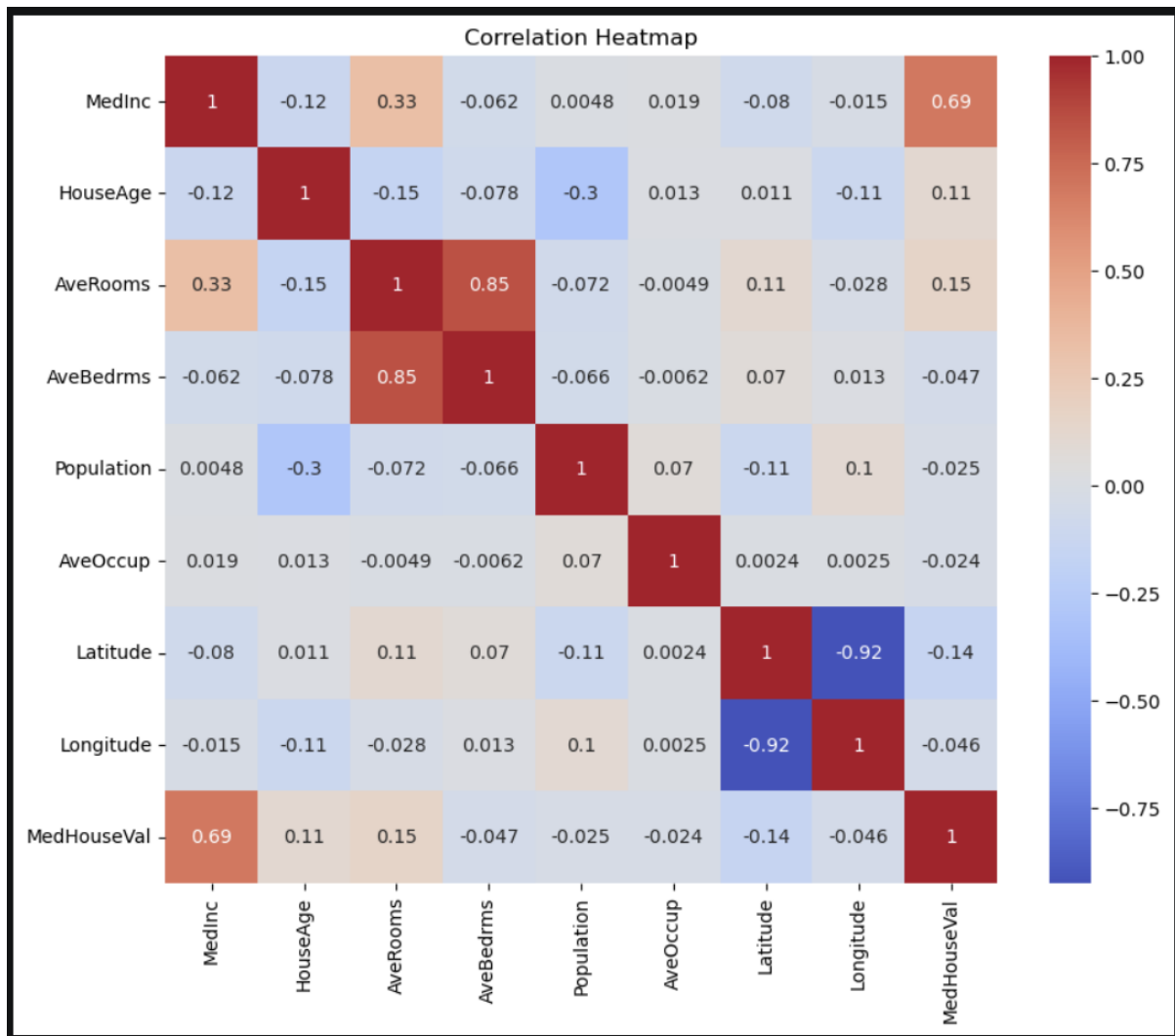
The following checks were performed:

- Dataset shape
- Data types of columns
- Statistical summary
- Missing value analysis
- Correlation analysis

The dataset was found to be clean and did not contain any missing values.

Observation

The correlation heatmap showed that Median Income (MedInc) has the strongest positive relationship with house prices. This indicates that areas with higher income generally have higher house values.



4. Data Preparation

The dataset was divided into:

- Training Data (80%)
- Testing Data (20%)

The training data was used to train the model, while the testing data was used to evaluate its performance on unseen data.

5. Model Building

A Linear Regression model from Scikit-Learn was used.

Linear Regression attempts to establish a linear relationship between the input features and the target variable.

The model was trained using the training dataset and then used to predict house prices on the testing dataset.

6. Model Evaluation

To evaluate model performance, the following metrics were used:

Mean Absolute Error (MAE)

MAE measures the average prediction error.

Root Mean Squared Error (RMSE)

RMSE gives more importance to larger errors and indicates overall prediction accuracy.

R^2 Score

R^2 Score indicates how well the model explains the variation in house prices.

Results

Metric	Value
MAE	0.533
RMSE	0.746
R^2 Score	0.576

The obtained R^2 score indicates that the model is able to explain a significant portion of the variation in house prices.

```
mae = mean_absolute_error(y_test, y_pred)

rmse = np.sqrt(mean_squared_error(y_test, y_pred))

r2 = r2_score(y_test, y_pred)

print("Mean Absolute Error (MAE):", round(mae, 3))
print("Root Mean Squared Error (RMSE):", round(rmse, 3))
print("R2 Score:", round(r2, 3))
```

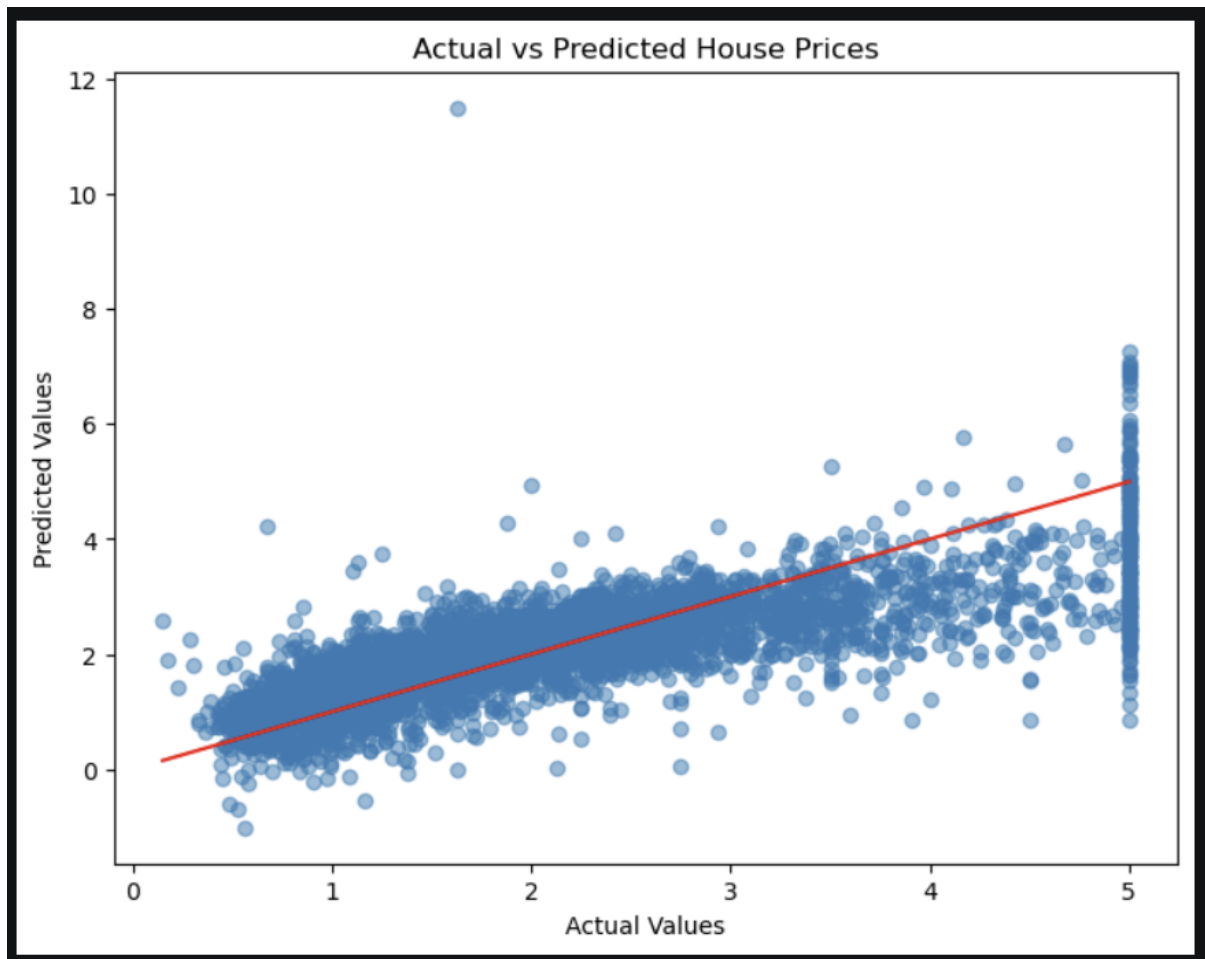
✓ 0.0s

```
Mean Absolute Error (MAE): 0.533
Root Mean Squared Error (RMSE): 0.746
R2 Score: 0.576
```

7. Actual vs Predicted Values

A scatter plot was generated to compare actual house prices with predicted house prices.

If the predictions are accurate, most points should lie close to the diagonal reference line.



Observation

Most points were concentrated around the reference line, indicating that the model learned the general relationship between housing features and house prices.

However, some deviations were observed for very high-value houses, suggesting that the model may not perfectly capture all complex relationships present in the dataset.

8. Challenges Faced

During this project, understanding the dataset and interpreting evaluation metrics were the main challenges. Learning how different features influence house prices also required careful analysis.

9. Future Improvements

The performance of the model can be improved by:

- Applying feature engineering techniques
- Removing outliers

- Using data scaling
- Trying advanced regression algorithms such as:
 - Random Forest Regressor
 - Gradient Boosting Regressor
 - XGBoost Regressor

10. Conclusion

In this project, a Linear Regression model was successfully developed for house price prediction using the California Housing Dataset. The complete machine learning workflow including data exploration, preprocessing, model training, prediction, and evaluation was implemented successfully.

The project provided practical experience with Scikit-Learn, data visualization, and performance evaluation techniques. It also demonstrated how machine learning can be applied to solve real-world prediction problems.